

Divya Natekar

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Education

New York University, Tandon School of Engineering

Brooklyn, NY

Master of Science - Urban Data Science, GPA: 3.58/4.0

Sept 2024 - May 2026

◦ Courses: Data Analysis, Machine Learning, Virtual & Augmented Reality, Artificial Intelligence, Advanced Spatial Analytics, Urban Computing & Artificial Intelligence

Manipal University Jaipur

Jaipur, India

Bachelor of Technology - Computer Science & Engineering / Minor in Cloud Computing Applications, GPA: 7.7/10 Jul 2019 - Jul 2023

◦ Courses: Big Data Analytics, Computer System Architecture, User Interface Design, Deep Neural Networks, Cloud Computing Applications

Sondervick College

Veldhoven, Netherlands

Cultural Exchange Program - Europe Meets India (EUMIND)

Jul 2015

Research Interests

Urban Artificial Intelligence; Agent-Based Modeling; Retail Mobility Modeling; Human Mobility Analysis; Spatial Data Science; LiDAR Point Cloud Processing; Explainable Urban Models; Large Language Models for Urban Analytics

Research

Thinking on the Move: Agent-Based Retail Simulation for Downtown Brooklyn

Sept 2025 - May 2026

NYU Center for Urban Science + Progress & Downtown Brooklyn Partnership

(Project Sponsor: Mark Landolina, Senior Director, Real Estate and Economic Development, Downtown Brooklyn Partnership; Mentor: Prof. Takahiro Yabe, Assistant Professor, Department of Technology Management and Innovation and CUSP, NYU Tandon)

- **Agent-Based Urban Simulation:** Developed an agent-based retail simulation framework for analyzing lunchtime mobility and restaurant choice behavior in Downtown Brooklyn.
- **LLM-Enhanced Retail Modeling:** Integrated large language models, gravity-based retail models, and spatial interaction methods to model qualitative dining preferences beyond distance-based assumptions.
- **Urban Data Integration:** Worked with mobility datasets, restaurant review embeddings, and urban POI data to evaluate underserved retail corridors and consumer behavior patterns.
- **Planning Decision Support:** Supported stakeholder-oriented retail planning and intervention analysis through predictive urban simulations.

Transient Object Removal from Urban LiDAR Point Clouds

Jun 2025 - Aug 2025

NYU CUSP Summer Guided Research

(Mentor: Prof. Debra Laefer)

- **Machine Learning Workflow:** Designed and implemented filtering workflows to remove transient urban objects such as pedestrians and vehicles from LiDAR point cloud scans.
- **3D Urban Modeling:** Improved fidelity and consistency of 3D urban models using spatial filtering and point-cloud processing techniques.
- **Research Recognition:** Selected for the NYU Tandon CUSP Experiential Learning Scholarship for Summer 2025 guided research.
- **Presentation:** Presented research findings at the BATWorks Climate Event and NYU CUSP Fall 2025 Research Showcase.

Urban Wildfire Risk Prediction using Remote Sensing and Deep Learning

2026

NYU Center for Urban Science + Progress

(Mentor: Prof. Rishabh Chauhan)

- **Remote Sensing Pipeline:** Developed an urban wildfire risk prediction pipeline using MODIS satellite imagery, NASA FIRMS wildfire datasets, and deep learning methodologies.
- **Model Development:** Implemented machine learning workflows in Python and Google Colab using Hugging Face and PyTorch-based architectures.
- **Fine-Tuning Exploration:** Explored parameter-efficient fine-tuning methods including LoRA and PEFT for wildfire classification and prediction tasks.
- **Spatial Visualization:** Generated spatial visualizations and geospatial outputs for urban environmental risk assessment.

Projects

Mech-On-Wheels

2020

Google Startup Weekend Honorable Mention

- Developed a prototype emergency roadside-assistance application connecting users with nearby mechanics during vehicle breakdowns.
- Designed a ride-hailing-inspired workflow to reduce response time and operational inefficiencies during vehicle emergencies.

Experience

L&T Technology Services (LTTS)

Airoli, India

Software Development Intern

Jan 2023 - Jul 2023

- **Backend Development:** Built consumer and backend components for a proof-of-concept healthcare data platform integrating Connected Wearable Devices with the Philips HSDP cloud ecosystem.
- **Real-Time Data Engineering:** Developed real-time data ingestion and processing pipelines using Java Spring Boot, Apache Kafka, and Spring Vault.
- **Cloud Infrastructure:** Provisioned and managed cloud infrastructure using Terraform and Microsoft Azure services.
- **DevOps & APIs:** Implemented CI/CD workflows through Azure DevOps, performed API testing with Postman, and documented services using Swagger within Agile development workflows.
- **Data Analysis:** Conducted data analysis and visualization workflows in Python to generate insights from healthcare datasets.

Leadership & Involvement

IEEE Student Branch, Manipal University Jaipur

Jaipur, India

Senior Coordinator

Jun 2020 - Jun 2021

- Contributed to the IEEE Delhi Section and supported initiatives through the IEEE Women in Engineering Club.
- Assisted in organizing workshops, seminars, and outreach events promoting diversity and participation in STEM disciplines.

Randomize(); Computing Club, Manipal University Jaipur

Jaipur, India

Head of Programs

Jun 2020 - Jun 2021

- Led planning and execution of technical seminars and programming events for the university computing community.
- Coordinated a seminar on “Python Workload in Azure” in collaboration with Microsoft MVP Karthikeyan VK and the PyJaipur community.

Additional

Programming & Data Science: Python, Java, C, C++, SQL, JavaScript, HTML, CSS, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, XGBoost, Matplotlib, Power BI

Urban Analytics & Geospatial Computing: GIS, QGIS, Spatial Data Analysis, Agent-Based Modeling, LiDAR Processing, CloudCompare, 3D Modeling, Mobility Data Analytics

Artificial Intelligence & Machine Learning: Large Language Models, Deep Learning, Clustering, Time Series Analysis, Remote Sensing, Computer Vision, Explainable Urban AI

Cloud & Software Engineering: Microsoft Azure, Azure DevOps, Terraform, Docker, Google Cloud Platform, Java Spring Boot, Apache Kafka, Git, Postman, Swagger

Certifications: Google Cloud Certifications: Baseline ML & AI, Infrastructure, Networking Fundamentals, Cloud Security

References

Debra Laefer: Professor at the NYU Tandon School of Engineering and the Center for Urban Science + Progress, New York University.

Takahiro Yabe: Assistant Professor at the Department of Technology Management and Innovation and the Center for Urban Science + Progress, New York University.

Rishabh Chauhan: Industry Assistant Professor at the Center for Urban Science + Progress, New York University.